Tianhao (Lawrence) Qian

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Research Interests: Reliable & Efficient AI, LLM Agents & Post-Training, Weak-to-Strong Transfer, Model Intervention, Game Theory

Education

Purdue University — West Lafayette, IN
Incoming M.S. Student in Electrical and Computer Engineering (ECE) — *Spring 2027*

Southeast University (SEU) | **Major GPA: 87.5/100** — Nanjing, China
B.S. in Applied Mathematics; Minor in Computer Science — *June 2026*

- **Selected coursework:** Topology (100), Stochastic Processes (96), Machine Learning (96), and Linear Algebra (92).

University of California, Berkeley — Berkeley, CA
Visiting Student, EECS Department; selected for the CDSS Research Track — *Fall 2025*

- **Graduate coursework:** Foundations of Decisions, Learning, and Games (A-); Natural Language Processing (A-).

Research Assistant

The University of Hong Kong — Hong Kong
Research Assistant, LLM Agents | *Supervisor: Jiayu Chen* — *Sep. 2026 – Present*

- Research on reliable LLM-agent search and coding-agent post-training, including certified path pruning and temporal credit assignment under limited compute.

The Chinese University of Hong Kong, Shenzhen — Remote
Research Assistant | *Supervisor: Lixu Wang* — *May 2026 – Present*

- Research on machine unlearning, intervention localization, and weak-to-strong model mapping across differently scaled language models.

Ongoing Research

Decision-Guided Relative Temporal Credit and Causal Repair (DG-RTC). Develop a decision-oriented framework for temporal credit assignment in multi-step agents, combining cross-turn attribution, cost-aware counterfactual evaluation, and verified causal repair to improve post-training decisions.

Cross-Model Functional Causal Support (CFCS). Investigate transferable functional correspondences across language models, using causal evidence from smaller models to identify intervention locations in larger models without independently reconstructing their full intervention maps.

OpenPAC: Certificate-Gated Learning for LLM Agents. Develop certificate-gated supervision to prevent LLM agents from learning from unverified or poisoned feedback, combining executable evidence with clean–adversarial paired training.

From Deliberation to Instinct. Investigate how expensive test-time deliberation can be amortized into a learned, state-dependent computation policy that decides when to search, inspect, test, call tools, or stop.

Publications

1. PAC-CF: Calibrating Irreversible Frontier Pruning in LLM-Guided Search — *2026*
T. Qian, J. Chen, Z. Sun, L. Wang | Manuscript Under Review

- **Problem + insight:** Introduced *trustworthy search-tree pruning* with PAC-CF, calibrating a score-gap threshold from held-out native search traces to preserve a verifier-valid continuation despite biased LLM evaluations.
- **Evidence:** Across 3 domains and 5 controllers, improved ToolTree full-test utility by **+4.38 points** at B100; natural-termination runs used **18.95% fewer physical requests** and **23.76% fewer tokens**.

2. Relative Kinetic Utility: Calibrating Cross-Layer Credit for Global Structured LLM Pruning — *2026*
T. Qian, G. Qi, J. Chen | Manuscript Under Review

- **Problem + insight:** Introduced a *label-free cross-layer credit signal* using final-state activation–gradient participation and block-relative normalization to compare channels fairly under a global sparsity budget.
- **Evidence:** At **40% sparsity** on Qwen2.5-7B, beat GISP by **+3.79 Mean3 points** under questions-only calibration; relative normalization contributed **+14.42 Mean3 points** in ablation.

3. Storage Is Not Strategy: State-Conditioned Support Control for LLM Unlearning — *2026*
T. Qian, Z. Hong, C. Gao, K. Chen, L. Wang | Manuscript Under Review

- **Problem + insight:** Demonstrated that locating stored knowledge does not identify the best weights to edit; introduced *intervention-value scoring* and calibrated, state-conditioned re-ranking during unlearning.
- **Evidence:** Storage localization reached **AUROC 0.981**, yet agreed with the better intervention on only **17/36** targets; terminal utility improved in **all 6** LACUNA NPO/SimNPO comparisons (NPO gains **+0.431 to +0.848**).

Publications (Continued)

4. **T. Qian**, Z. Li, J. Cao, X. Shi, and L. Rutkowski. "Alternating Gradient Flow Utility: A Unified Metric for Structural Pruning and Dynamic Routing in Deep Networks." 2025.
   - **Problem + insight:** Designed a gradient-flow utility signal to preserve functional pathways under extreme compression and decoupled offline structural selection from inexpensive online routing.
   - **Evidence:** Avoided structural collapse at **75% compression** on ImageNet-1K; reduced heavy-expert use by approximately **50%** on ImageNet-100 without sacrificing full-model accuracy.
5. **T. Qian**, G. Qi, Z. Y. Wu, R. Gu, X. Liu, and C. Lyu. "Large Language Model for Discrete Optimization Problems: Evaluation and Step-by-step Reasoning." 2025.
   - **Finding:** Benchmarked LLM reasoning across differently scaled discrete optimization tasks; observed that chain-of-thought prompting was **not consistently beneficial**, highlighting task- and model-dependent reasoning limits.
6. **T. Qian**, Z. Li, J. Cao, X. Shi, and L. Rutkowski. "Resource-constrained Amazons chess decision framework integrating large language models and graph attention." 2024.
   - **Problem + insight:** Combined graph-attention filtering and Monte Carlo Tree Search to learn strategic decisions from imperfect LLM-generated supervision under limited compute.
   - **Evidence:** Achieved a **66.5% win rate** against GPT-4o-mini at a 50-node search budget, with **15–56%** decision-accuracy gains over evaluated baselines.
7. J. Li, Y. Meng, L. Wang, **T. Qian**, S. Zhai, and G. Qi. "Taxonomy Induction Using LLMs: An Enhanced Framework by Integrating Doubly-Checked Mechanism and Self-Evaluation Strategy." *CCKS-IJCKG 2024*, accepted.
   - Contributed to iterative checking and refinement of LLM-generated taxonomy candidates for more reliable hierarchical knowledge induction.

Academic Service

Invited Reviewer, *Journal of Artificial Intelligence Research (JAIR)*, 2026.
Faculty-supervised Review Contributor, *International Conference on Learning Representations (ICLR 2026).*

Professional Experience

United Technology Computer Co., Ltd. Feb. 2025 – July 2025
AI Engineering Intern
- Fine-tuned language-model agents with LoRA on mixed task datasets and developed LangChain-based API workflows for model orchestration, request handling, and evaluation.

Gray Heron Intelligence Mar. 2024 – June 2024
NLP Intern
- Engineered and maintained a retrieval-augmented generation system, including vector retrieval, document indexing, and context integration for domain-specific knowledge bases.

Selected Honors

Honorable Mention, COMAP Mathematical Contest in Modeling (MCM), 2024 | **Provincial First Prize**, Zhengda Cup National College Students Market Survey and Analysis Competition, 2024
Second-Class Scholarship, Strong Foundation Program, Southeast University, 2024 and 2025.

Technical Skills

Programming and systems: Python, Bash, Linux, Git, LaTeX. **ML:** PyTorch, Hugging Face Transformers, PEFT/LoRA, LangChain, LLM APIs.
Methods: LLM-guided search, coding-agent post-training, RL credit assignment, weak-to-strong transfer, Monte Carlo Tree Search, statistical decision theory, structured pruning, machine unlearning, mathematical optimization, algorithmic game theory.